MILITARY SPECIFICATION

PANS, DRAIN, AUTOMOTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and two sizes of drain pans used for collecting oil being drained from automotive vehicles.

1.2 Classification.

1.2.1 Sizes. The drain pans shall be of the following minimum sizes, as specified (see 6.2.1):

Size I -- four gallon capacity
Size II -- six gallon capacity

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-P-291 - Paperboard, Wrapping and Cushioning.
PPP-T-60 - Tape, Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more drain pans for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

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* 3.2 Design. The drain pan shall be new and of the manufacturer's latest design meeting the requirements specified herein. The pan shall be a shallow receptacle equipped with an anti-splash baffle for pouring. Each pan shall include handles for lifting, pouring, and positioning the pan under an automotive vehicle to receive oil drainage. The design of non-metallic pans shall include ribbed bottoms and sides to provide strength and rigidity.

* 3.2.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3 Construction. All parts and materials used to manufacture the pan shall be free of defects. Patching processes shall not be used to reclaim defective parts or surfaces. Surfaces shall be clean and free of extraneous materials. Edges shall be rounded or beveled.

* 3.4 Welding, brazing, and soldering. Welding, brazing, and soldering, when applicable, shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.5 Performance and product characteristics. Drain pans shall be capable of receiving and conveying drained engine oil without splashing or leakage when filled to the capacities specified in 3.5.1. The drain pan shall incorporate provisions which shall allow one operator, in a standing position, to push or pull the pan on a shop floor. Pans shall also include lifting and pouring capabilities.

3.5.1 Capacities. The size I drain pan shall have a capacity of no less than four gallons and the size II pan shall have a capacity of no less than six gallons.

3.5.2 Leakage. Drain pans shall be capable of retaining a solution of nonflammable liquid, having a viscosity equivalent to that of gasoline, without leakage after being pulled a distance of 20 feet on a concrete floor at a normal walking pace.

3.5.3 Pan height. The height of the size I drain pan shall be no greater than five inches and that of the size II pan no greater than 5 1/2 inches.

3.5.4 Shock. The drain pan shall be capable of withstanding without damage the shock of being dropped three times on its bottom surface on a concrete floor from a vertical distance of one foot.

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3.6 Components.

* 3.6.1 Pan body. When constructed of metallic material, the sheet metal shall be no less than Manufacturer's Standard Gage No. 26 thickness (0.018 inch). When constructed of oil resistant plastic material, the thickness of the material shall assure that the completed pan meets the requirements of this specification. Body joints and seams, when applicable, shall be leakproof and shall comply with the requirements of 3.5 thru 3.5.4. The body shall incorporate a pouring spout designed to provide controlled pouring of the pan's contents.

* 3.6.2 Baffles. The drain pan shall include a baffle to prevent oil from running over the outer edge of the pan during pouring operations.

* 3.6.3 Handles. The drain pan shall be furnished with two full hand grip handles to assist in lifting and carrying the pan and to permit pushing and pulling the drain pan on level concrete when filled to capacity. One handle shall be located opposite the pouring spout. The other handle shall in no way hinder pouring and shall be capable of being rotated at least 180 degrees from the hinged point to facilitate tilting of the drain pan.

* 3.7 Corrosion resistance. Metallic components shall be inherently corrosion resistant or shall be plated or treated to resist corrosion in accordance with standard commercial practice.

3.8 Product identification. The drain pans shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.9 Workmanship. Standards of workmanship shall assure that the drain pans shall have the stability, strength, and characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Leakage test (see 4.6.1).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance shall be in accordance with MIL-STD-105 using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination (4.5)	II	1.0
Leakage test (4.6.1)	II	1.0
Packaging inspection (4.7)	S-4	4.0

* 4.5 Examination. Visually and manually examine the pans to determine conformance with the requirements of 3.2, 3.2.1, 3.3, 3.4, 3.5.3, and 3.6.1 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the pan has passed the examinations prescribed by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

* 4.6.1 Leakage test. The drain pans shall be filled to capacity with the solution specified in 3.5.2 and then pulled a distance of 20 feet on a concrete floor at a normal walking pace. The floor shall approximate the surface and condition of a concrete automotive shop floor. After being pulled, the pans shall be permitted to stand, filled with the liquid solution, for 15 minutes. Upon conclusion of that time period, the pans shall be examined for leakage.

4.6.2 Shock test. The drain pans shall be dropped three times on their bottom surfces to a concrete floor from a vertical height of one foot. On conclusion of the test, the drain pans shall be examined for evidence of damage and then tested in accordance with 4.6.1. There shall be no evidence of leakage or of damage affecting use.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or Industrial. Unless otherwise specified (see 6.2.1), Industrial preservation shall be provided. When

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military (Level A) preservation is required, it shall be so specified in the contract.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Cleaning and drying of the drain pans shall be in accordance with any applicable process and procedure of MIL-P-116.

* 5.1.1.2 Preservative application. Non-corrosive resistant surfaces of metallic drain pans (see 3.7), shall be coated with type P-19 preservative of MIL-P-116.

5.1.1.3 Unit packaging. The unit method of protection shall conform to method I of MIL-P-116 except that method III of MIL-P-116 shall be used for those pans which are inherently corrosion resistant or treated to be corrosion resistant. Each pan shall be wrapped and cushioned with paperboard conforming to PPP-P-291, any type and style. The wraps shall be secured with tape, any type, conforming to PPP-T-60.

* 5.1.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

* 5.2 Packing. Packing shall be Level A, Level B, or Industrial. Unless otherwise specified (see 6.2.1), Industrial packing shall be provided. When military (Levels A and B) packing is required, it shall be so specified in the contract.

5.2.1 Level A. A quantity of unit packaged drain pans (see 5.1.1.3) shall be packed in an exterior shipping container conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The quantity of pans and the gross weight of the container shall be governed by the limitations of the container specification. Closure and strapping of the container shall be in accordance with the applicable box specification. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

* 5.2.2 Level B. Drain pans shall be packed as specified in 5.2.1 except shipping containers shall conform to PPP-B-640, class 2.

* 5.2.3 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

* 5.3 Marking. Marking shall be Military or Industrial. Unless otherwise specified (see 6.2.1), Industrial marking shall be provided. When military marking is required, it shall be so specified in the contract.

* 5.3.1 Military marking. Military marking shall be as specified in MIL-STD-129.

* 5.3.2 Industrial marking. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The drain pans covered by this specification are intended to be used when draining oil from automotive engines.

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6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2.1).
(c) First article when required (see 3.1).
(d) Product identification if different (see 3.8).
(e) First article inspection if different (see 4.3).
(f) Quality conformance inspection if different (see 4.4).
(g) Sampling for quality conformance inspection if different (see 4.4.1).
(h) Level of preservation, packing, and marking if different (see 5.1, 5.2, and 5.3).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-45819C	Pans, Drain, Automotive

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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